# RapidBoxForest: Link-Envelope-Guided Box Forests for Fast Multi-Query Manipulation

TIAN, Yu and Ren, Hongliang

***Abstract*—Fast repeated-query manipulation benefits from reusable volumetric planning structure, but high-quality IRIS-style convex regions are expensive to construct and rebuild. This paper presents RapidBoxForest (RBF), a link-envelope-guided box-forest planner for fast multi-query manipulation. The central geometric primitive maps endpoint interval envelopes to link interval envelopes via capsule Minkowski decomposition and convex-hull generator bounding applied to *$C$-space box validation and multi-query reuse*: endpoint interval envelopes bound rounded-link endpoint motion over a joint box, and a radius-lifted link envelope provides an inexpensive filter for box validation and collision rejection. A rapidly-exploring random tree (RRT)-inspired grower builds a scene-specific box forest into a queryable corridor graph, and a Lifelong Envelope Cache Tree (LECT) selectively caches kinematics-keyed envelope records when replay is cheaper than recomputation. On the reported benchmark set, the d18 cache prewarm stores 524287 evidence records in 20.22 s and reopens in 1.460 s with 262144 reused leaf records. On Shelf+IIWA, the current LinkIAABB warm rows replay 3420 external records in E2 and 7201 in E3 while preserving 1/1 and 5/5 strict-audit success; wall time remains about 1.8 s and 1.6 s because scene-specific growth and validation dominate this inexpensive envelope setting. A dynamic-update study further shows 1.9×–140× speedups over warm rebuild on matched obstacle-edit paths. This paper keeps the original planning formulation and title, but limits its quantitative claims to results regenerated from the current RapidBoxForest implementation.**

***Index Terms*—Motion planning, multi-query planning, link interval envelopes, box-region planning, final audit.**

## I. Introduction

Motion planning for high-degree-of-freedom (DOF) manipulators becomes especially demanding when a system must answer many related queries rather than one isolated instance. In warehouses, packing cells, and shared workcells, robot kinematics stay fixed while obstacles, task endpoints, or local scene edits change. The central question is how to construct reusable $C$-space planning structure quickly enough that repeated queries can benefit from region-level search without paying a large scene-specific preprocessing cost.

Iterative Regional Inflation by Semidefinite Programming (IRIS) and Graph-of-Convex-Sets (GCS) methods show why free-region construction is valuable: once good convex regions are available, graph search or convex optimization can produce high-quality motions over those regions [1], [2], [3]. The limitation is construction cost. In high-dimensional manipulator scenes, inflating and validating high-quality convex regions can dominate the planning budget and must often be repeated when the scene changes. Sampling-based planners such as probabilistic roadmap (PRM) and rapidly-exploring random tree (RRT) methods avoid this region-construction cost [4], [5], [6], [7], but their reusable structures are usually zero-volume graphs, so repeated queries still pay for local collision checking and do not obtain a reusable free-region cover.

RBF targets a different design point by constructing simpler box regions quickly. Instead of inflating general polytopes, it grows axis-aligned boxes in configuration space, connects accepted boxes into a graph, and answers later queries by searching the resulting box corridor. This trades per-region expressivity and sometimes path length for cheap construction, simple adjacency, and direct reuse across many queries. The intended claim is a system-level one: a fast box-region planning pipeline can be a useful final-audited alternative when multi-query latency matters more than obtaining the shortest path from a more expensive region decomposition.

The key technical ingredient is a link interval envelope computed from endpoint interval envelopes. For rounded link models, RBF reduces link motion over a joint box to the motion of a zero-radius centerline plus a final radius lift. Endpoint interval envelopes bound the endpoint trajectories over the joint box; link interval envelopes then enclose the swept rounded link and can be stored as axis-aligned bounding boxes (AABBs), $k$-DOP slabs, or SupportHull records. Strict interval sources provide an optional conservative validation path, while tighter practical sources are used as fast filters and discharged by the same final scene audit used for all reported planners.

On top of this geometric primitive, the grower borrows the rapid frontier-expansion idea of RRT: samples choose useful frontiers, a local region oracle materializes a free box around the projected seed, and accepted boxes extend a connected box forest. The result is also an adaptive cell-decomposition view of manipulator $C$-space, but one biased toward the regions needed by the current repeated-query workload. A Lifelong Envelope Cache Tree

The authors are with the Department of Electronic Engineering, The Chinese University of Hong Kong (CUHK), Hong Kong SAR, China. E-mail: ty1997@link.cuhk.edu.hk.

(LECT) then memoizes scene-independent endpoint and link-envelope evidence keyed by robot kinematics and joint intervals; it is an acceleration layer for repeated materialization, not a separate planning claim.

The evaluation therefore asks what the current implementation can support under the adopted protocol. The main axes are cache prewarm and replay cost, Shelf+IIWA cold-versus-warm build and query time, strict-audit success, envelope-representation trade-offs across the link-envelope families including staged AABB/KDOP26/SupportHull cascades, and dynamic update speedup over warm rebuild. Broader cross-planner baselines and full multi-robot generalization remain future work.

We therefore position the paper as a planning-systems contribution enabled by a geometry primitive. The endpoint-to-link envelope is the main geometric idea; the method shows how that primitive supports a reproducible low-build-cost multi-query design point under final audit.

**The paper's main contribution is a reproducible final-audited multi-query planning system and its measured design point. It is supported by three concrete elements:**

1) *An endpoint-to-link envelope primitive for $C$-space box validation.* Capsule Minkowski decomposition and convex-hull generator bounding are individually classical facts already combined in articulated-body CCD work [8], [9], [10]. RBF applies this two-step pipeline to *$C$-space box growth and multi-query reuse*: a switchable endpoint source family—AA-backed interval forward kinematics (IFK), hierarchical IFK (HIFK), CritSample, analytical extrema—compact per-representation link records (AABB, KDOP26, SupportHull) for staged filtering, and a kinematic-keyed LECT cache for cross-scene envelope reuse. No prior work has been found applying this pipeline to $C$-space box validation or multi-query planning reuse (sections III and VI-C).
2) *A self-contained link-envelope-guided box-forest planner.* Scene-specific $C$-space box forests are grown through seeded frontier expansion, consolidated into a connected corridor graph, and queried by graph search and local refinement. LECT provides an optional kinematic-keyed cache for repeated envelope materialization across builds (sections IV, V and VI-C).
3) *An implementation-aligned evaluation.* Shelf+IIWA, cache-mechanism, envelope sweep, and dynamic-update studies are evaluated under the strict audit protocol adopted in this paper. The reported rows are generated from the current implementation and do not carry forward the older cross-planner comparison tables (sections VI-A, VI-C and VI-D).

The paper's evidence chain is: Section III develops the endpoint and link interval-envelope primitive for fast box validation; section IV describes the cache optimization that reuses expensive envelope records; section V turns link-envelope-filtered boxes into a rapidly expanding box-forest planner; and section VI evaluates the resulting build/query, cache-reuse, and dynamic-update behavior under the adopted protocol.

## II. Related Work

### A. Convex Region Planning and Graphs of Convex Sets

Convex region planners approach motion planning by constructing a certified cover of free space and then solving graph search or convex optimization on top of that cover. IRIS introduced the now-standard idea of computing large collision-free convex regions by alternating separating hyperplanes and region inflation [1]. Later work extended the same line into configuration space, improved region quality, or improved graph compactness through visibility-graph and clique-cover constructions [11], [12], [13]. Marcucci *et al.* made the Graph-of-Convex-Sets formulation particularly influential by showing that motion planning can be cast as optimization over a graph of certified convex regions [2], [3].

RBF is aligned with this family in objective but not in geometric primitive: it uses axis-aligned boxes in $C$-space rather than general polytopes, trading per-region expressivity for lower build overhead and simpler adjacency bookkeeping. The resulting cover is coarser than a high-quality convex decomposition, but it avoids repeated hyperplane-based region inflation and is easier to rebuild under repeated-query workloads. The comparison in this paper therefore concerns fast reusable free-region construction versus tighter, more scene-coupled regional convexification, rather than box and polytope representations in isolation.

### B. Swept Volume Approximation and Capsule Geometry

Computing the swept volume of a moving rigid body has been studied extensively in computational geometry and computer-aided manufacturing (CAM) communities [14], [15]. For a convex body $P = \mathrm{conv}(v_1, \ldots, v_m)$, bounding each generator trajectory by a conservative outer set $\widehat{\Gamma}_i(I)$ yields

$$\mathcal{S}_I(P) \subseteq \mathrm{conv}\Big(\bigcup_i \widehat{\Gamma}_i(I)\Big),$$

a bound that is implicit in oriented bounding box (OBB) tree construction for swept mesh primitives [16].

For capsule links this bound reduces to bounding only the two segment endpoints. The Proximity Query Package (PQP) exploits exactly this Minkowski decomposition ($C = S \oplus B_2(r)$, sweep of $C$ reduces to sweep of $S$ plus radius lift) [17]; GJK and its robust implementation by van den Bergen apply the same identity for distance queries [18], [19]; the Flexible Collision Library (FCL) generalises the swept-sphere interface to a full collision library [20].

### C. Interval Forward Kinematics and the CCD Endpoint-Bounding Pipeline

Interval arithmetic [21], [22] provides inclusion-monotone enclosures of real-valued functions over

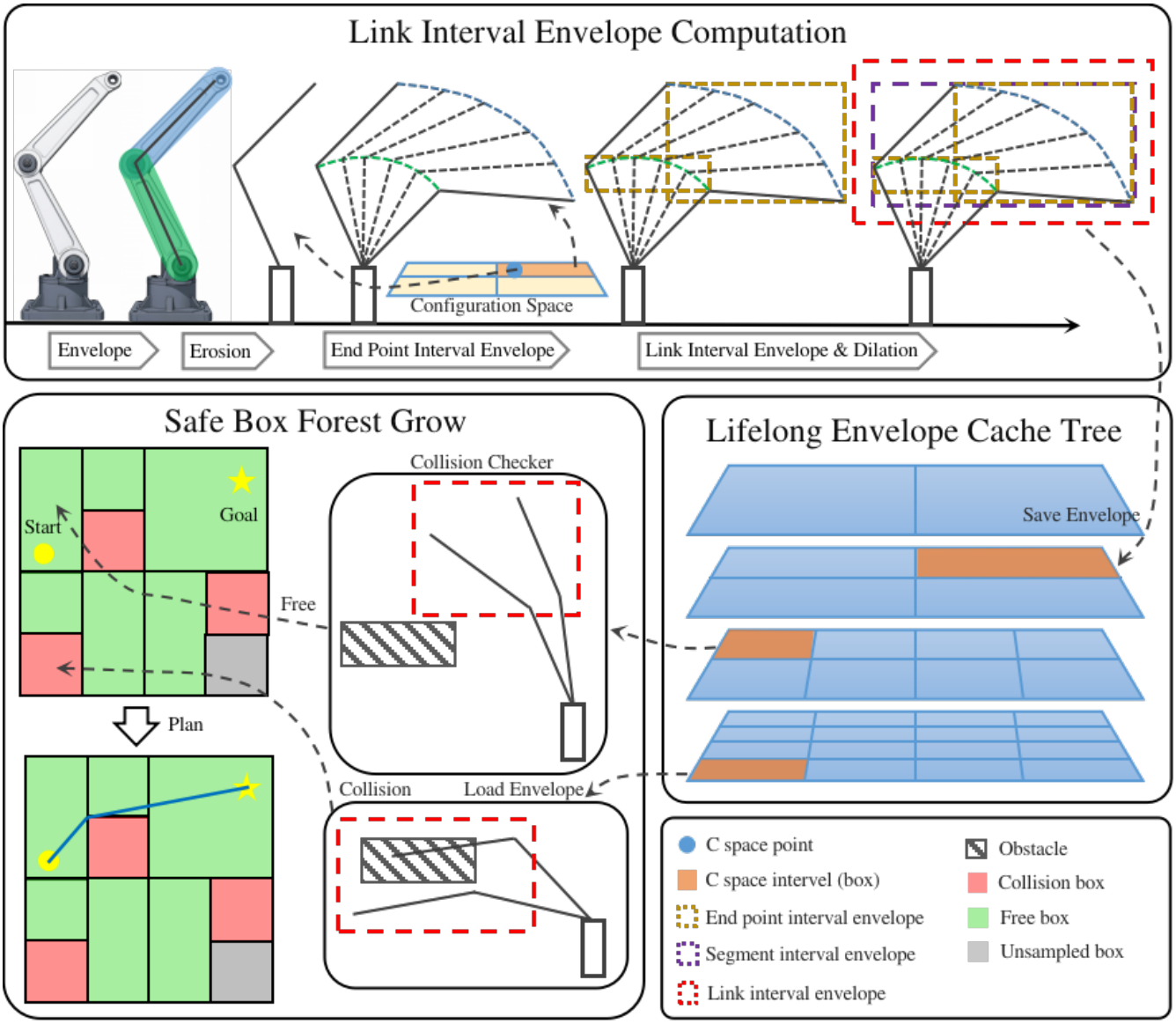



Fig. 1. RBF pipeline for link-envelope-guided multi-query planning. Top: endpoint interval envelopes induce link interval envelopes over a $C$-space box. Middle: LECT optionally reuses kinematics-keyed endpoint and link-envelope records across repeated builds. Bottom: scene filtering and an RRT-inspired grower expand a connected box-region forest for lightweight corridor queries and final audit.

parameter intervals. Snyder applied interval arithmetic to parametric geometry to compute conservative bounding boxes for curves and surfaces without mesh sampling [23]. Merlet extended DH-chain interval propagation to certified workspace analysis for parallel and serial manipulators [24], [25].

These ideas meet directly in the continuous collision detection (CCD) literature for articulated bodies. Redon, Kim, Lin, and Manocha [8] model each robot limb as a line-swept sphere (LSS, i.e. capsule) and propagate interval Denavit-Hartenberg (DH) transforms along the kinematic chain to obtain *interval vectors bounding the two LSS endpoint positions* over the motion time step; the swept capsule is then bounded by padding the endpoint bounding box with the link radius. This is explicitly the two-step pipeline

$$\textit{interval FK endpoint bounding} \longrightarrow \mathrm{bbox}([\mathbf{p}_i] \cup [\mathbf{p}_{i+1}]) \oplus B_2(r) = \textit{link swept-capsule bound}$$

that is the geometric core of RBF. The journal extension by Redon, Lin, Manocha, and Kim [9] applies the same framework to general robot arms and kinematic trees; that work has accumulated over 240 citations and is a standard CCD reference. Zhang *et al.* [10] replace interval arithmetic with Taylor models in the same pipeline to reduce wrapping errors, analogous to how RBF's CritSample and analytical sources tighten the IFK bound. Schwarzer *et al.* [26] address a multi-query motivation similar to RBF's—reusing collision evidence across planning calls—but at the path-segment level rather than as volumetric $C$-space records. Corrales *et al.* [27] apply sphere-swept line (SSL) bounding volumes dynamically for human–robot safety monitoring using the same structural observation that bounding a capsule's motion reduces to bounding its two endpoints.

### D. Novelty Boundary: Prior CCD vs. RBF

The two-step endpoint-interval-to-swept-capsule pipeline is therefore unambiguously present in the CCD literature for time-step motion. The differences that motivate the RBF contribution claim are: (a) the parameter domain is a *C-space joint-box interval* rather than a trajectory time step, so the goal is to classify whether *any* configuration in the box is collision-free, not whether a single given path collides; (b) the endpoint source is a switchable interface—IFK (conservative), CritSample (practical), and analytical extrema (tightest)—rather than a fixed algorithm, allowing the conservative vs. performance trade-off to be controlled without changing the link-representation or planner layers; (c) compact per-representation link records (AABB, KDOP26, SupportHull) are organised for staged filtering during box growth, a knob absent from CCD libraries; and (d) the endpoint and link envelope records are organised in LECT, a persistent *kinematic-keyed* cache that reuses evidence across multi-query builds in related scenes, a reuse level not addressed by CCD libraries or path-segment caches. No prior work has been found that applies the two-step endpoint bounding pipeline to $C$-space box growth or to multi-query volumetric planning reuse. RBF is complementary to the CCD stack: link envelopes reduce repeated geometric work on candidate boxes, while discrete scene checks remain the final authority for the reported paths.

### E. Multi-Query Reuse and Adaptive Configuration-Space Decomposition

Multi-query planners typically amortize effort by reusing a graph structure such as a roadmap. PRM is the classical example: build a connectivity graph once and reuse it for many start-goal pairs [6], [28]. LazyPRM, lazy edge checking, and certificate-based planners postpone or cache edge validation so that online search does not repeatedly evaluate the same discrete local connections [29], [30]. Experience-based and trajectory-library systems reuse prior solutions or local motion snippets, often as initialization or bias for a new query rather than as a conservative decomposition of a region of configuration space [31].

RBF operates at a different reuse level. It grows and reuses volumetric boxes rather than only paths, binary collision outcomes, or zero-volume edges. PRM-like methods reuse connectivity and must revalidate affected edges after scene changes. Convex-region planners reuse higher-quality regions, but those regions are usually coupled to the obstacle layout in which they were constructed.

The method also inherits the central idea of adaptive cell decomposition and subdivision planning: represent free space by cells, split cells where the current evidence is insufficient, and search the adjacency graph induced by the accepted cells [32], [33]. The difference is the operating regime. Classical cell-decomposition examples are often low-dimensional workspace or explicit $C$-space constructions, whereas RBF applies the same adaptive-cell idea to high-dimensional manipulator $C$-space through a link-envelope oracle. Scene-specific obstacle filtering is layered on top of reusable kinematic envelope evidence, with LECT available only as a cache for expensive records. The trade-off is explicit: RBF accepts axis-aligned boxes and over-approximation in exchange for simple validation, adjacency maintenance, and repeated-query throughput.

### F. Sampling-Based Planning and Trajectory Optimization

Sampling-based planners remain standard baselines when the goal is to solve a single query quickly. RRT-Connect is particularly effective in manipulator planning because bidirectional growth often succeeds with very little global structure [34]. Asymptotically optimal and informed variants such as RRT*, Batch Informed Trees (BIT*), and Fast Marching Tree (FMT*) improve path quality and search efficiency when more time is available [7], [35], [36], [37]. The Open Motion Planning Library (OMPL) has made these methods easy to deploy and benchmark in a common software stack [38], [39]. However, standard RRT variants produce single-use zero-volume evidence and typically restart from scratch on the next query.

By contrast, RBF uses rapidly exploring, adaptive expansion to sample and materialize $C$-space boxes from the envelope oracle, producing a volumetric corridor structure that can be queried repeatedly within the same environment. Trajectory optimizers address a different stage of the pipeline. CHOMP, STOMP, TrajOpt, and Gaussian-process motion planners refine an initial path by local descent or sequential convex optimization [40], [41], [42], [43]. RBF uses these methods as complementary points of comparison: its aim is to provide a reusable volumetric backbone on which graph search and local post-processing can operate.

## III. Link Interval Envelopes

This section develops the main geometric primitive used by RBF: a map from a manipulator joint box to compact link interval envelopes. The goal is to make box validation cheap enough for repeated-query planning. A strict interval source gives a conservative validation option, while tighter sampled or support-based variants are practical filters whose reported paths are discharged by the common final audit. Thus the strongest free-region statement in this section is conditional on the conservative source and on paths remaining inside the validated box union; the main planning experiments report final-audited performance separately. The construction first bounds endpoint trajectories and then applies a constant-radius lift for rounded link geometry.

Here $\mathcal{Q}$ denotes the joint-limit box explored by RBF, $I \subseteq \mathcal{Q}$ one joint interval, and $\mathcal{O} \subset \mathbb{R}^3$ the obstacle set. Endpoint envelope $E_k(I)$ outer-bounds endpoint $k$, and link envelope $B_\ell(I)$ outer-bounds the swept volume of link $\ell$ over the same interval. Together they define the

reusable planning primitive $I \mapsto \{E_k(I)\} \mapsto B_\ell(I)$. When the endpoint source is itself conservative, the resulting link envelope is conservative as well; when a tighter empirical source is used, the same representation serves as a fast broad- or mid-phase filter before final scene auditing rather than as a standalone certificate.

### A. Geometric Foundation

Let $T(q)x = R(q)x + t(q)$ denote the rigid transform induced by a configuration $q \in I$. For a reference body $C_0 \subset \mathbb{R}^3$, its pose at $q$ is $C(q) = T(q)C_0$. We write $\mathcal{S}_I(C_0)$ for the exact swept workspace set over $I$ and $\mathcal{C}_I(C_0)$ for its convex envelope. RBF usually works with the convex envelope because the downstream representations are AABBs, hulls, or fixed-direction slab bounds. The following construction combines two classical computational-geometry facts—capsule Minkowski decomposition and convex-hull generator bounding—in a new planning context. Both facts are used individually in swept-sphere collision libraries and in CCD work for articulated bodies [8], [9]; the novelty here is applying their combination to *C-space box validation and multi-query reuse* rather than to single-query time-step CCD.

*Minkowski sum separation (radius isolation).* For a capsule $C = S \oplus B_2(r)$, rigid motion commutes with the fixed Euclidean ball, giving exact two-stage sweep bounds:

$$\mathcal{S}_I(C) = \mathcal{S}_I(S) \oplus B_2(r), \qquad \mathcal{C}_I(C) = \mathcal{C}_I(S) \oplus B_2(r).$$

RBF therefore bounds the zero-radius seed first and applies the link radius at the end. This identity is exact for capsules; for sharp polytopes it requires the shape to be a rounded offset body.

*Convex-hull generator bound.* For a convex body $P = \mathrm{conv}(v_1, \ldots, v_m)$, bounding each vertex trajectory by an outer set $\widehat{\Gamma}_i(I)$ suffices to bound the swept polytope:

$$\mathcal{S}_I(P) \subseteq \mathrm{conv}\Big(\bigcup_i \widehat{\Gamma}_i(I)\Big).$$

For a capsule link the seed is the center segment $[a, b]$, so only two endpoint trajectories need to be bounded.

Combining these two facts yields the practical recipe. For a capsule link $[a, b] \oplus B_2(r)$, RBF bounds only the two endpoint trajectories, forms their convex hull, and applies the radius lift:

$$\widehat{\mathcal{C}}_I([a, b] \oplus B_2(r)) = \mathrm{conv}\big(\widehat{\Gamma}_a(I) \cup \widehat{\Gamma}_b(I)\big) \oplus B_2(r).$$

The numerical task reduces to computing reusable envelopes for two endpoint trajectories per link, fully determined by endpoint-trajectory enclosures plus the chosen downstream representation.

**Proposition 1** (Endpoint envelopes induce link interval envelopes)**.** *Consider a rounded link $L = [a, b] \oplus B_2(r)$ and a joint interval $I$. If endpoint envelopes $E_a(I)$ and $E_b(I)$ satisfy $\Gamma_a(I) \subseteq E_a(I)$ and $\Gamma_b(I) \subseteq E_b(I)$, then*

$$\mathcal{S}_I(L) \subseteq \mathrm{conv}\big(E_a(I) \cup E_b(I)\big) \oplus B_2(r).$$

*Consequently, any outer representation of the right-hand side is a valid link interval envelope for the whole joint box $I$.*

*Proof.* For each $q \in I$, the center segment of the moved link is the convex hull of the two moved endpoints, so $T(q)[a, b] \subseteq \mathrm{conv}(\Gamma_a(I) \cup \Gamma_b(I))$ over the entire interval. The endpoint inclusions place this set inside $\mathrm{conv}(E_a(I) \cup E_b(I))$. Adding the fixed radius ball commutes with the rigid-motion sweep of the capsule, giving the stated containment. □

### B. Endpoint Interval Envelopes

Figure 1 illustrates the AABB extraction pipeline. Endpoint interval axis-aligned bounding boxes (iAABBs) are computed by interval forward kinematics (IFK) or tighter certified AA-based variants, combined into a conservative zero-radius centerline envelope, and padded by the link radius before collision querying. The remaining numerical task is to outer-bound each endpoint trajectory over $I$ by a reusable set $E_k(I)$. Endpoint envelopes are the common interface between kinematic propagation and the downstream link-envelope representations.

For endpoint $k$, let $\Gamma_k(I)$ denote its exact workspace trajectory over interval $I$. Any conservative set that contains $\Gamma_k(I)$ is an endpoint interval envelope. In the baseline implementation this envelope is simply an axis-aligned bounding box, because boxes are cheap to store, easy to combine, and directly reusable by all downstream link representations.

*a) Interval forward kinematics (IFK).:* In the public API used in this paper, `EndpointSource.IFK` is the spelling for the IFK_AA-backed certified endpoint source. It propagates affine-arithmetic workspace bounds along the DH chain and remains the cheapest certified endpoint source in the active paper track, but it can loosen on wide boxes or long chains because one evaluation still accumulates residual over-approximation across the full joint box.

*b) Hierarchical IFK (HIFK).:* HIFK further reduces wrapping by recursively bisecting the joint box and evaluating AA-FK at the leaves before hulling the child endpoint boxes. This is not the same as the unsplit IFK baseline: a true depth-1 HIFK evaluation already splits once and hulls two leaves, whereas IFK performs one unsplit AA-FK pass on the original box. The mechanism study in section VI-C reports two representative split depths, HIFK_3 and HIFK_5, to quantify how additional hierarchical depth trades runtime for tighter certified endpoint boxes.

*c) Incremental FK state reuse.:* Tree descent also carries a transient FK state: interval-valued prefix transforms $[\boldsymbol{T}^{0:0}], \ldots, [\boldsymbol{T}^{0:n}]$ and the per-joint DH interval matrices used to build them. The full prefix chain is computed once at the root or after a restart; when only $I_k$ changes, prefixes before $k$ remain valid and only the suffix is recomputed. This short-lived state is not the persistent cache itself; it only keeps endpoint materialization cheap before LECT decides whether to retain the evidence.

*d) Critical-sample envelopes.:* The second source estimates the endpoint box from carefully chosen boundary

and interior samples. In practice these samples include interval endpoints, turning-point candidates near multiples of $\pi/2$, and a few midpoint or near-boundary probes. This often captures extrema that IFK wraps away, at modest extra cost. The limitation is completeness: separable per-joint samples do not certify coupled extrema. CritSample is therefore a performance-oriented source whose reported planning paths are discharged by the final scene audit unless an external certificate is added.

*e) Analytical extrema.:* The third source solves endpoint-coordinate extrema from the trigonometric structure of the chain. Because each revolute-joint transform is affine in $(\cos q_j, \sin q_j)$, an endpoint coordinate becomes a simple sinusoid when one joint is free and a low-degree bilinear trigonometric polynomial when two joints remain free. This yields a structured solver: vertices and 1D boundary edges can be enumerated exactly, 2D faces reduced to polynomial root finding, and higher-dimensional residual cases handled by local descent with interval verification or branch-and-bound closure. Analytical bounds are the tightest of the three sources, but also the most expensive, so they are better suited to offline certification and profiling than to the online path.

Taken together, these sources define a clear spectrum: IFK is the cheapest certified AA baseline, HIFK_3 and HIFK_5 offer progressively tighter certified outer bounds at increasing cost, CritSample is the cheapest tighter advisory bound, and the analytical source is the tightest but most expensive. Endpoint envelopes are the right intermediate abstraction: they are the state variable of the joint-box-to-envelope map, so the downstream link-envelope layer stays fixed while the upstream source changes with the certification and runtime budget.

### C. Link Interval Envelopes

Once endpoint envelopes are available, the remaining design question is how to represent the swept link volume compactly for box growth, collision filtering, and optional reuse. For a capsule link, RBF does not reason about the thick link directly. It first bounds the zero-radius center segment between the two endpoint envelopes and then pads that centerline envelope by the link radius. A conservative link interval envelope $B_\ell(I)$ is any set that contains the full swept capsule over interval $I$. In the default box-based implementation this is written as $B_\ell(I) = B^\circ_\ell(I) \oplus \mathcal{R}(r_\ell)$, where $B^\circ_\ell(I)$ bounds the centerline sweep and the padding operator $\mathcal{R}$ becomes $B_\infty$ for AABB envelopes.

The following representations share the same endpoint evidence but preserve different amounts of downstream geometric detail.

*a) AABB envelope.:* AABB envelopes are the simplest option. The ordinary LinkIAABB construction takes the two endpoint envelopes, wraps them in one box, and pads that box by the link radius:

$$B_\ell(I) = \mathrm{bbox}\big(E_{\ell-1}(I) \cup E_\ell(I)\big) \oplus B_\infty(r_\ell).$$

This representation can be tightened by subdividing the reference center segment into $S$ pieces, bounding each piece separately, and retaining the union of the padded sub-envelopes. Larger $S$ preserves more directional variation along the link while keeping reconstruction cheap once endpoint envelopes are available. The main caveat is that the subdivision must remain explicit: if all sub-envelopes are collapsed back into one global AABB, much of the gain disappears.

*b) KDOP slab envelope.:* A k-discrete-oriented-polytope ($k$-DOP) envelope stores lower and upper support values along a fixed set of directions. Here the 26-plane variant (KDOP26) uses the three coordinate axes, six face-diagonal axes, and four body-diagonal axes, for a total of 13 unsigned axes and 26 supporting planes. Endpoint boxes are projected onto these axes, padded by the link radius, and merged over each subsegment. Collision checking is correspondingly cheap: after the ordinary link AABB test, the slab test continues with the non-axis directions and exits on the first separating interval. The staged order first tests the DOP18 axes and only reaches the final DOP26 diagonals for near survivors.

*c) SupportHull envelope.:* When the endpoint envelopes are convex, their convex hull preserves directional structure that a single AABB discards. SupportHull is the compact runtime realization of that idea: instead of storing an explicit dense hull, it keeps the two endpoint AABBs and the link radius as a support-function record. For a query direction $d$, the support point is the better of the proximal and distal AABB support points, and collision refinement against an obstacle AABB uses a fixed-size Gilbert–Johnson–Keerthi (GJK) narrow phase with the obstacle inflated by the link radius and safety margin. This retains much of the convex- hull directional fidelity while keeping the default record compact; a hybrid variant may additionally retain KDOP26 intervals as a mid-phase before GJK, but that is treated as a separate runtime/storage point rather than as the default SupportHull envelope.

In this way, the link-envelope layer is separate from the upstream kinematics solver. Sections sections III-A to III-C define a modular endpoint-to-link envelope map from joint-space boxes to reusable workspace filters. Once endpoint envelopes are available, the remaining design choice is the representation and the corresponding planning-time cost/tightness trade-off.

## IV. Lifelong Envelope Cache Tree

The planner can materialize the envelopes of section III online. LECT is a secondary optimization for workloads in which repeated builds revisit similar joint boxes or use envelope representations expensive enough to replay. It stores scene-independent endpoint/link-envelope evidence keyed by robot kinematics and joint intervals. Scene-specific obstacle tests, box labels, graph connectivity, and final-audit outcomes remain outside the cache. This section therefore records only the cache interface used by the

planner; the scientific claim of the paper does not depend on LECT providing broad cross-scene transfer.

### A. Cache Key and Split Policy

LECT is a dynamically grown binary KD tree over the joint-limit box $\mathcal{Q}$. Each node represents one joint interval and stores interval bounds, split metadata, cached endpoint evidence from which link envelopes can be reconstructed, and residency state used by the implementation. The useful property is that the stored evidence is *kinematics-keyed* rather than *scene-keyed*: later scenes may replay a materialized envelope record, but they must rerun their own scene-local obstacle tests. The current storage format identifies nodes by 64-bit breadth-first heap ids (0 at the root and $2p+1, 2p+2$ for children). To keep this structural id scheme safe, the implementation enforces a configurable maximum tree depth $D_{\max} \leq 64$, with default $D_{\max} = 64$.

The split policy is envelope-aware rather than purely geometric. For a parent interval $I$, let $I_d^-$ and $I_d^+$ be the two children obtained by splitting joint dimension $d$ at its midpoint, and let $\mathcal{V}(I) := \sum_\ell \mathrm{vol}(B_\ell(I))$ be the cumulative downstream envelope volume. The default best-tighten score chooses the split with the smallest worst-child envelope burden,

$$d^* = \arg\min_d \max\left\{ \sum_\ell \mathrm{vol}(B_\ell(I_d^-)), \sum_\ell \mathrm{vol}(B_\ell(I_d^+)) \right\}.$$

The implementation uses this score in a depth-synchronous schedule so that all nodes at the same depth share one split decision. Candidate dimensions are also filtered by FK reachability: splitting a joint beyond the last active frame that contributes to any link endpoint cannot tighten the endpoint envelopes and is skipped.

### B. Runtime Use and Validation Boundary

LECT is queried by FINDFREEBOX, not by the graph search itself. Given a joint interval, the cache either replays endpoint evidence or materializes it online, then derives the requested link representation. Directional records are used as refinement evidence rather than as standalone free-space certificates: the runtime first evaluates padded link AABBs, then uses KDOP26 slabs or SupportHull/GJK only for survivors of the AABB test. When a node is split, child envelopes may refine the parent witness, but any compact parent record remains representation-dependent and is rechecked against the current obstacle set before a box can enter the forest.

### C. Storage and Warm-Build Semantics

Persistence is selective. Endpoint evidence is the common reusable record, but simple IFK+AABB evidence can be cheaper to recompute online than to reload. KDOP26 and SupportHull records are stored only when their tighter filters are expected to amortize storage and lookup cost. Implementation details such as node arrays and residency management are engineering choices at the implementation level; the paper does not claim runtime benefit from sector symmetry, because the cache mechanism operates in canonical-identity mode with an empty symmetry descriptor.

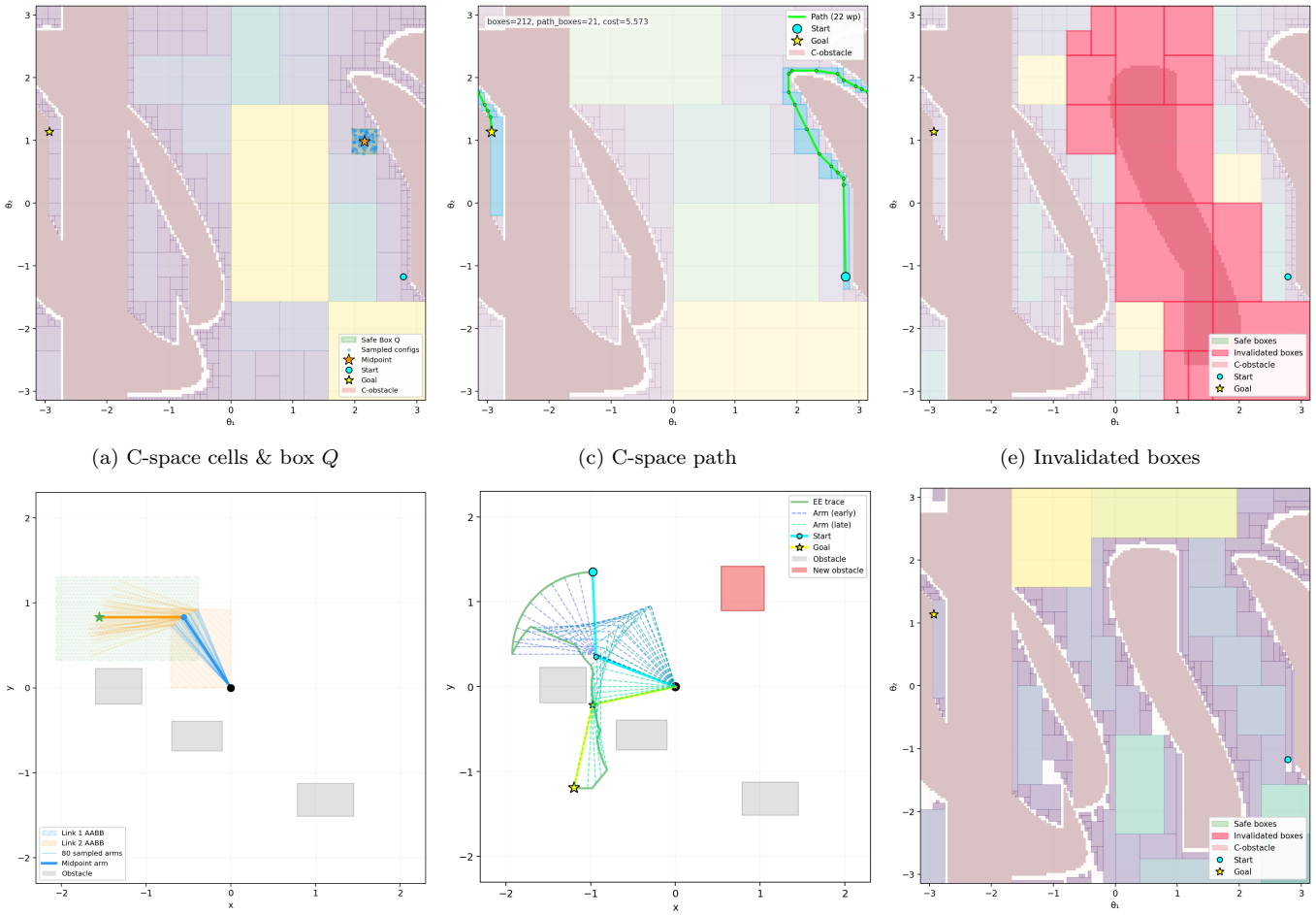
The prewarm phase follows the same separation. It materializes FK/link-envelope evidence only on the target leaf layer and reconstructs upper layers by child-hull propagation during checkpoint. This avoids charging direct FK at multiple depths for the same cache and keeps parent witnesses tied to the leaf records that generated them. Exact leaf evidence and propagated child-hull evidence are both treated as envelope records by the replay path; provenance does not create a separate reuse or validation mode.

In the warm-build path, LECT reuses only kinematic envelope evidence. On a cache hit, the current link-envelope representation is reconstructed from stored endpoint records and then revalidated against the current obstacle set. A warm build still regrows and revalidates scene-specific RBF boxes and never treats a persisted box label as a free-space certificate. In section V, LECT serves only as the materialization layer; obstacle checking, occupancy metadata, adjacency bookkeeping, and regrowth reactions to changing $\mathcal{O}$ remain inside the box-region forest routines. For the reviewer-sensitive Shelf+IIWA protocol, the external d18 cache is treated as read-only during evaluation: warm shelf rows disable active-database backfill and therefore cannot silently accumulate new online envelope evidence while being measured. The balanced random-scene study keeps that writeback path enabled because it measures cache growth on novel scenes rather than a fixed held-out shelf target.

## V. RapidBoxForest Construction

This section describes the planning layer of RBF. The algorithm grows a scene-specific forest of $C$-space boxes and exports the forest as a graph for repeated corridor queries. The growth loop is RRT-like, but the state added to the data structure is a free region rather than a point or edge: samples select frontiers, the link-envelope oracle materializes a usable box, and accepted boxes extend a connected region graph. In adaptive cell-decomposition language, the boxes are cells, FindFreeBox is the local refinement and classification primitive, and the forest graph is the adjacency structure used for query search. The link interval envelopes of section III and the optional LECT cache of section IV serve this planner as collision-filtering and materialization tools.

Throughout this section, a validated box means a box accepted by the selected scene-validation policy. In strict conservative mode, paths that remain inside the validated box union inherit the corridor statement below. In all reported planning experiments, paths that leave this union through repair, smoothing, or external composition are charged where used and passed through the same final scene audit.




(a) C-space cells & box $Q$ (c) C-space path (e) Invalidated boxes

(b) Workspace AABBs of $Q$ (d) Workspace path (f) After regrowth

Fig. 2. Illustrative RBF build, query, and obstacle-update cycle on a planar 2-link manipulator. (a, b) Adaptive $C$-space cells and workspace AABB envelopes of the highlighted box $Q$. (c, d) Box-corridor path in $C$-space and its workspace trajectory. (e, f) After obstacle insertion, intersecting boxes are invalidated (red), and localized repair/regrowth updates the explicit forest without requiring a full rebuild.

Conceptually, the build process has three stages. First, a seed-guided local oracle adaptively refines and classifies a $C$-space cell around a selected configuration. Second, repeated frontier expansion grows these cells into a connected forest, giving the method its region-valued RRT/PRM character. Third, consolidation coarsens redundant boxes before the structure is exported as a query graph. The consolidation pass is part of the reusable build phase and its time is included in reported build measurements.

### A. Construction Objective

Let $\mathcal{Q}$ denote the joint-limit box and let $\mathcal{O}$ denote the current obstacle set. The construction objective is to build a finite family

$$\mathcal{F} = \{B_1, \dots, B_N\}, \qquad B_i \subseteq \mathcal{Q},$$

such that every $B_i$ is a validated free-space box, the union $\bigcup_i B_i$ covers as much useful query-relevant free space as possible under the available box, time, and depth budgets, and the induced adjacency graph supports efficient downstream search. This object is scene-specific: changing $\mathcal{O}$ changes which intervals are accepted into the forest, even though the underlying envelope evidence in LECT remains reusable.

The central structural rule is that the forest is not merely a free-space cover, but a *corridor-compatible* free-space cover. Boxes are inserted under a connected-box constraint.

Viewed as adaptive cell decomposition, this objective is intentionally workload-biased rather than globally exhaustive. The planner does not try to classify every cell in $\mathcal{Q}$. It refines and accepts cells near useful frontiers, task seeds, and connector candidates, then keeps the resulting adjacency graph as the reusable planning object.

*a) Connected-box rule.:* For boxes

$$B_i = \prod_{d=1}^{n} [l_i^{(d)}, u_i^{(d)}], \qquad B_j = \prod_{d=1}^{n} [l_j^{(d)}, u_j^{(d)}],$$

RBF treats them as connected when, in every coordinate, the two intervals either overlap or are separated by at most a small tolerance $\varepsilon_c$:

$$\max\{l_i^{(d)}, l_j^{(d)}\} \le \min\{u_i^{(d)}, u_j^{(d)}\} + \varepsilon_c, \qquad d = 1, \dots, n.$$

When $\varepsilon_c = 0$, this reduces to ordinary interval intersection or boundary touching in every dimension. A small positive

**Algorithm 1** FindFreeBox: local box materialization around a seed.

**Input:** Seed configuration $q$, LECT $T$, obstacle set $\mathcal{O}$, depth cap $D_{\max}$
**Output:** Validated interval $I_v$ or failure

$v \leftarrow \mathrm{root}(T)$; $d \leftarrow 0$; initialize current interval and FK state
**while** $d \le D_{\max}$ **do**
  *Cache materialization*
  materialize the endpoint and link envelopes of $v$ if needed
  *Scene validation*
  **if** $v$ is marked fully occupied **then**
    **return** failure
  **end if**
  **if** $v$ is validated collision-free with respect to $\mathcal{O}$ **then**
    **return** interval $I_v$
  **end if**
  *Refinement*
  **if** $d = D_{\max}$ **then**
    **return** failure
  **end if**
  **if** $v$ has no children **then**
    split $v$ along the depth-synchronous split dimension of depth $d$
  **end if**
  $k \leftarrow$ split dimension of $v$
  $v \leftarrow$ the child of $v$ whose interval contains $q$; update the current interval
  update the FK state from dimension $k$ onward; $d \leftarrow d + 1$
**end while**
**return** failure

tolerance absorbs the boundary offset used by grower seeds and floating-point roundoff.

This rule is deliberately weaker than requiring a full shared face. It treats overlap, touching, and tolerance-scale gaps uniformly as graph-connectivity evidence, while validity remains attached to the accepted box union itself. The output of the construction stage is a forest of validated boxes whose graph connectivity is aligned with the corridor object queried later, without imposing a stricter face-sharing invariant than the grower actually maintains.

### *B. Find Free Box*

FINDFREEBOX descends LECT from the root toward the leaf containing seed $q \in \mathcal{Q}$, materializing endpoint and link envelopes lazily along the path, skipping saturated subtrees via occupancy metadata, and returning the current interval as a validated box candidate as soon as the scene oracle accepts it. Tree topology and endpoint records are reused from the scene-independent cache, while scene-specific validation is deferred to each grower visit. Algorithm 1 gives the complete pseudocode.

The termination behavior is straightforward. Under depth cap $D_{\max}$, each loop iteration either returns immediately or descends one more level in the tree, so the procedure can make at most $D_{\max} + 1$ decisions. In strict conservative mode, any returned interval has passed envelope disjointness against $\mathcal{O}$; in performance-oriented modes, the returned interval is a validated planning region that remains subject to the common final audit used for reported paths.

### *C. Forest Growth*

Starting from an initial seed set, the forest is grown by repeatedly invoking the find-free-box procedure on carefully chosen candidate configurations. Each seed first produces a root box through the find-free-box procedure, and these root boxes initialize one or more connected components of the forest. The main grower should be understood as an evidence-guided rapidly exploring process rather than as a generic random cover: every iteration chooses a frontier component, proposes a nearby interval worth testing, and accepts the result only when it extends the forest by a usable connected corridor segment.

Each iteration makes two explicit heuristic decisions: it samples a target configuration $q_t$, then chooses a boundary seed on an existing box face that is closest to that target. The reported planning configuration uses a sequential Bernoulli schedule with component-connector probability $p_c = 0.45$, query-root bias $p_g = 0.20$, intertree-root bias $p_i = 0.25$, and unexplored-volume probability $p_u = 0.45$. A component-connector target between disconnected root components is attempted first. If no connector target is used, coordinated multi-tree mode may target a root from another active component; because $p_i \le 0.5$ in the reported runs, no high-bias pulse is active and the sustained cap is also 0.25. If no intertree target is used, the grower then tries a query-root target, then a LECT unexplored-volume leaf, and finally a uniform root-interval sample. The same reported configuration uses 5000 boxes, a 60 s build timeout, an FFB depth cap of 120, eight worker threads, a component-connector candidate limit of four, a staged component-connect radius of 0.35 in normalized $L_\infty$ distance, a post-connect time budget of 450 ms, and a quality floor of 64 connected boxes. These numbers are planner settings rather than envelope-theory parameters, but they are part of the self-contained specification for the reported rows.

Given a target $q_t$ and an existing box $B = \prod_j [\ell_j, u_j]$, the grower enumerates feasible faces $f = (B, d, s)$, where $d$ is a joint dimension and $s \in \{-1, +1\}$ is the lower or upper side. Let $\tau$ be the adjacency tolerance and $\epsilon_b = \max(\epsilon_{\mathrm{guard}}, \tau/4)$. A face is feasible only if stepping outside it by $\epsilon_b$ remains inside the global root interval. In the reported configurations $\epsilon_{\mathrm{guard}} = 0$, so this formula uses the $\tau/4$ floor. The seed associated with the face is

$$q_f[j] = \begin{cases} u_d + \epsilon_b, & j = d,\ s = +1, \\ \ell_d - \epsilon_b, & j = d,\ s = -1, \\ \mathrm{clip}(q_t[j], \ell_j + \epsilon_b, u_j - \epsilon_b), & j \neq d, \end{cases} \tag{1}$$

with clipping to $[\ell_j, u_j]$ when a box is thinner than $2\epsilon_b$. Faces are ranked by the squared target distance

$$\sigma(f; q_t) = \|q_t - q_f\|_2^2, \tag{2}$$

and the implementation retains the 128 lowest-score face candidates across the scanned frontier boxes. It then tries them in increasing score order, skipping a candidate if the corresponding LECT leaf is temporarily cooled or if a component-connection cache has already recorded the same boundary seed. The first remaining candidate becomes the seed for FINDFREEBOX.

The candidate box is accepted only if FINDFREEBOX

**Algorithm 2** GrowForest: budgeted expansion of a connected box-region forest.

**Input:** LECT $T$, obstacle set $\mathcal{O}$, seed set $\mathcal{S}$, box budget $N_{\max}$, time budget $t_{\max}$
**Output:** Validated box-region forest $\mathcal{F}$

$\mathcal{F} \leftarrow \emptyset$; initialize connected components from $\mathcal{S}$
**for all** $q \in \mathcal{S}$ **do**
  $B \leftarrow \text{FindFreeBox}(q, T, \mathcal{O})$
  **if** $B \neq \emptyset$ **then**
    insert $B$ into $\mathcal{F}$ as a root box
  **end if**
**end for**
**while** $|\mathcal{F}| < N_{\max}$ and time budget not exhausted **do**
  draw target $q_t$: connector with prob. $p_c$, query/intertree root with prob. $p_g$ or $p_i$, unexplored LECT leaf with prob. $p_u$, otherwise uniform
  enumerate feasible faces $f = (B, d, s)$ of frontier boxes and compute $q_f$ and $\sigma(f; q_t)$
  choose the first of the 128 lowest-score face seeds that is not cooled and not in the frontier-seed cache
  set $(q_{\text{seed}}, B_{\text{near}}) \leftarrow (q_f, B)$ for that face
  $B_{\text{new}} \leftarrow \text{FindFreeBox}(q_{\text{seed}}, T, \mathcal{O})$
  **if** $B_{\text{new}} \neq \emptyset$ and $\text{BoxesConnected}(B_{\text{new}}, B_{\text{near}}, \tau)$ **then**
    insert $B_{\text{new}}$ into $\mathcal{F}$; $\text{UpdateComponents}(\mathcal{F})$
  **else**
    record the failed LECT leaf; cool it after one failed attempt for a 300-box horizon
  **end if**
**end while**
**return** $\mathcal{F}$

validates it and the new box is connected to its parent under the same adjacency predicate used by the query graph: all dimensions overlap up to tolerance $\tau$, and at least one dimension touches within tolerance or all dimensions overlap. Failures are also handled explicitly. When the LECT leaf containing a failed seed reaches the hard-frontier failure threshold of one failed attempt, the leaf is skipped for a horizon of 300 subsequently accepted boxes, unless a later depth stage raises the active search depth and enables a retry. Cooling and the frontier-seed cache do not alter the validation boundary; they only reduce repeated calls on already failed or already covered frontier regions. The resulting method is summarized in algorithm 2.

The main reported configuration uses coordinated multi-goal growth. Five seed configurations initialize the shelf roots, and the grower uses the fixed resource budget reported in section VI-A. This choice keeps the growth schedule simple enough to preserve one authoritative forest state while still exposing reusable LECT evidence across threads and cache-warm reruns. Staged wavefront growth and task-batched multi-threaded growth are treated as engineering variants, and the parallel-scaling experiment reports the present limit of the task-batched variant. The output of this section is the query-ready box-region object only.

### *D. Forest Consolidation*

Raw growth tends to produce a forest that is finer than necessary. The purpose of consolidation is to replace a large collection of locally valid boxes by a smaller collection of equally valid but more useful regions. Two operations are central.

**Algorithm 3** ConsolidateForest: graph-preserving promotion, merging, and pruning.

**Input:** Validated forest $\mathcal{F}$, LECT $T$, obstacle set $\mathcal{O}$
**Output:** Consolidated validated forest $\mathcal{F}'$

$\mathcal{F}' \leftarrow \mathcal{F}$
*Stage 1: sibling promotion*
**repeat**
  promote any validated sibling leaves whose parent is collision-free
**until** no eligible sibling pair remains
*Stage 2: validated connected/contact merging*
merge exact connected pairs whose union is itself a single box
**for all** connected candidate pairs $(B_i, B_j)$ in $\mathcal{F}'$ **do**
  build a candidate parent region $H$ and score its volume expansion
  **if** the chosen envelope representation of $H$ is collision-free with respect to $\mathcal{O}$ **then**
    replace $B_i, B_j$ by $H$
  **end if**
**end for**
*Stage 3: containment pruning*
remove only contained boxes whose adjacency and segment witnesses transfer to selected covering regions
**return** $\mathcal{F}'$

The first is *promotion*. When two occupied sibling leaves are both collision-free and their parent interval is also collision-free, the two leaves can be replaced by the parent. This is the most local form of coarsening and directly reuses the multiresolution structure of LECT.

The second is *multi-level coarsening*. Exact connected merges are the cleanest case: if two boxes are connected and their union is itself one box, the merged region can be accepted directly after validation. More general connected candidate merges are handled by constructing a candidate parent region and accepting it only when its chosen envelope representation remains collision-free against the current obstacle set. This is where the representation choices of Sections III and IV become operational. In an AABB-based forest, the compact parent witness is typically the enclosing AABB of the two child boxes. KDOP parents can merge child slab intervals by per-axis minima and maxima, while SupportHull parents are kept compact only when the merged endpoint-hull witness remains acceptable; otherwise the child witnesses are kept. Greedy non-sibling candidates are ranked by a volume-expansion score, so exact face merges are preferred and high-inflation parents are deferred unless they pass validation under the scene oracle. The consolidation logic is summarized in Algorithm 3. The final containment-pruning step removes only boxes that are fully covered by accepted validated regions and whose graph incidence can be transferred to a selected covering region. It changes neither the represented free-space union nor the corridor witnesses used by query search.

Validity follows directly: every accepted promotion or merge has itself passed the scene oracle, so the represented free-space union cannot shrink. Containment pruning removes only boxes fully covered by a selected region that inherits their adjacency witnesses, preserving all corridor connectivity.

### E. Corridor Query

After growth and consolidation, the forest is exported as a graph $G = (V, E)$ with one vertex per validated box. Two vertices are connected by an edge when their boxes satisfy the connected-box rule above. Lightweight querying is possible because this connected-box structure is enforced during growth and consolidation rather than repaired after graph export.

If disconnected adjacency islands remain after growth, an optional connector stage attempts targeted bidirectional sampling bridges between nearest frontier boxes of different components. A successful bridge is stored as a segment edge rather than as a claim that the two boxes overlap. This distinction is important for both the RBF query layer and the GCS negative controls: a segment edge is a collision-checked path witness between two boxes, whereas a GCS edge over convex regions requires feasible overlap of the regions themselves.

This pair $(\mathcal{F}, G)$ is the final planning object produced by the build stage. Querying it is intentionally lightweight. Given start and goal configurations $q_s, q_g$, the planner associates them with start-side and goal-side boxes in $\mathcal{F}$ and then searches $G$ for a connecting box sequence. The query problem is reduced to box-corridor retrieval inside an already constructed free-space representation.

A shortest path on $G$ yields a box sequence whose union defines a corridor between the two query endpoints. A geometric path is then recovered as a waypoint chain inside that corridor. In the experiments, this candidate is then passed through the same OMPL-style simplification/local post-processing and fixed-step final audit used by the comparison pipelines. That post-processing can move the final reported path outside the source box union. Consequently, the box-corridor statement below applies only to contained paths before or after post-processing, while the reported planning success in the main curves is the common fixed-resolution audit outcome.

**Theorem 1** (Conditional validated box-corridor path)**.** *Let $\mathcal{F}$ be a forest whose boxes have each passed conservative link-envelope validation against obstacle set $\mathcal{O}$. If a continuous path $\gamma : [0, 1] \to \mathcal{Q}$ satisfies $\gamma([0, 1]) \subseteq \bigcup_{B \in \mathcal{F}} B$, then every configuration on $\gamma$ is collision-free with respect to $\mathcal{O}$.*

*Proof.* For any $s \in [0, 1]$, the containment assumption gives a validated box $B \in \mathcal{F}$ with $\gamma(s) \in B$. By construction, validation of $B$ means every conservative link envelope for $B$ is disjoint from $\mathcal{O}$. Because each such envelope contains the link geometry for every configuration inside $B$ (section III), the robot geometry at $\gamma(s)$ is also disjoint from $\mathcal{O}$. Since $s$ was arbitrary, the whole path is collision-free. □

*a) Reporting protocol.:* Theorem 1 is a conditional statement: it covers path portions contained in a conservatively validated box union. It is not the acceptance rule for the main experimental comparisons. When OMPL simplification, local repair, smoothing, or external composition leaves that union, the path is accepted only if it passes the same fixed-resolution final scene collision audit used for the baselines. The main performance curves should therefore be read as post-processed, fixed-audit planning results rather than as evidence that every reported path satisfies the box-corridor containment hypothesis.

For dynamic scenes, the same explicit forest representation supports local graph maintenance without compulsory rebuilds. Each accepted box records the obstacle evidence that blocked or validated its envelope tests; obstacle deletions cannot invalidate existing boxes accepted under the same validation policy, because free space only grows. Deletion triggers only a spatial dirty-region pass around the removed obstacle to choose local regrowth seeds, after which any new box is validated by the same oracle as in the original build. Obstacle insertion is stricter: the new obstacle is inflated by the dirty-region padding, all overlapping boxes are rechecked exactly, invalid boxes and incident graph edges are removed, and every surviving segment edge is re-audited in the edited scene. The repair stage then regrows from deleted-box centers, live neighbours, original task seeds, and component-boundary candidates; each committed box must pass the active validation policy before it enters the forest. Finally, query failure can trigger a bounded local repair search, again filtered by the same final audit. Warm rebuild remains available as a measured baseline and fallback policy.

If no contained conservative corridor is recovered, no box-corridor guarantee attaches to that query. The construction still exports a concrete box-region planning object, and section VI evaluates its build cost, query cost, path quality, and success rate after the same post-processing and fixed-resolution final audit used for the other planners.

## VI. Experiments

This section evaluates RBF as a multi-query planning system under the current evaluation protocol. Returned paths are compared only after the implemented query pipeline has completed its own post-processing and strict final audit.

*a) Scope.:* The main evidence consists of the Gate 0 configuration smoke, the E5 lifelong-cache mechanism study, the E1 envelope sweep, the E2/E3 Shelf+IIWA cold-versus-warm rows, and the dynamic-update capability comparison. A warm random-scene study across IIWA, UR5, and Panda is also reported for completeness.

*b) Reporting Protocol.:* Planner success in this paper means strict path-audit success under the current query pipeline. Cache reuse is allowed only at the envelope-materialization layer: target builds still regrow and revalidate scene-specific boxes, and warm rows are never credited for persisted free-space labels. For the reviewer-sensitive Shelf+IIWA rows, the external d18 cache is read-only during evaluation and active-database backfill is disabled.

*c) Fixed Configuration.:* All rows use the configuration `rbf_ifk_aa_aafkvol_d40_s15_canonical_lifelong`: the public API spelling `EndpointSource.IFK` for

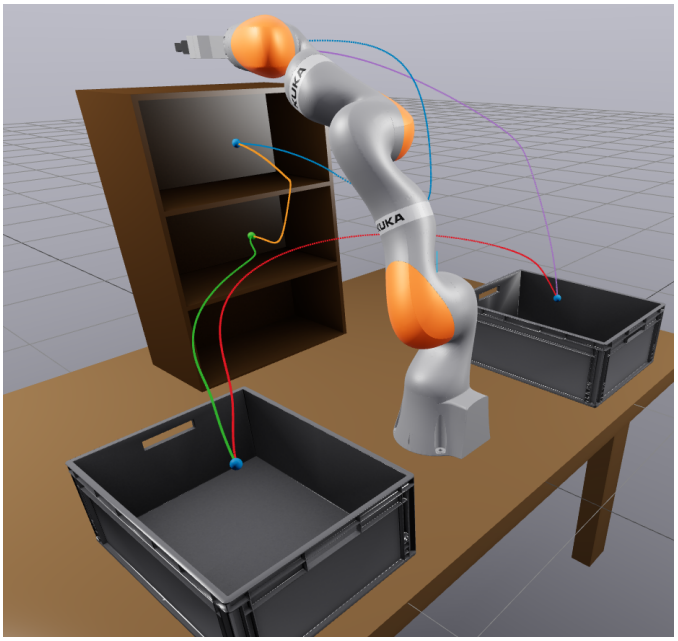


Fig. 3. Canonical Shelf+IIWA scene used in the experiments. The E3 main run evaluates five fixed query pairs on this scene under the strict audit protocol.

the IFK_AA-backed certified endpoint source, an AAFKVolumeMin split schedule through depth 40, find-free-box start depth 15, strict-certificate validation with certified-only commit, a depth-18 prewarm, a default LECT tree-depth cap of 64, and the adopted link-envelope family {LinkIAABB, KDOP26, SupportHull}.

*d) Platform and Scenes.:* Planner rows were generated with one worker thread; the standalone link-envelope representation microbenchmark uses the batch parallelism recorded in its result artifact. Shelf+IIWA uses the canonical Marcucci scene with five fixed query pairs. The random-scene benchmark uses balanced easy/medium/hard scenes for IIWA, UR5, and Panda at five scene seeds per setting.

### A. Shelf+IIWA Efficiency and Path Quality

Figure 3 and table I summarize the Shelf+IIWA rows. The single-query E2 shelf row remains strict-audit safe in both cold and warm modes: wall time changes from 1.820 s to 1.812 s, and the warm row replays 3420 external evidence records while both rows use one bounded repair. The five-query E3 main row also preserves all strict audits: cold and warm wall times are 1.587 s and 1.588 s, respectively, with the warm row replaying 7201 external evidence records and requiring no query repairs.

The current LinkIAABB shelf rows therefore demonstrate correct external-evidence replay without persisted free-space labels, not a large wall-time speedup. This is expected for this inexpensive representation: scene-specific box growth, obstacle validation, and query post-processing dominate the measured wall time once endpoint evidence replay is enabled.

### B. Balanced Random-Scene Pipeline Study

The multi-robot random-scene benchmark reports the same fixed protocol on IIWA, UR5, and Panda across easy, medium, and hard settings. IIWA and Panda remain strict-audit safe across the three difficulties, while UR5 is the limiting case in the present data.

### C. Mechanism Studies: Envelopes and Cache

TABLE II
Width-wise CritSample link-envelope comparison under the same $S = 4$ short-link split, including staged AABB/KDOP26/SupportHull cascades.

| Width | Envelope | $V$ (m$^3$) | Payload | Eval ($\mu$s) |
|---|---|---|---|---|
| 0.02 | LinkIAABB | 0.0017 | 384.0 B | 0.5 |
| 0.02 | KDOP26 | 0.0017 | 1.62 KB | 1.4 |
| 0.02 | SupportHull | 0.0017 | 832.0 B | 1.5 |
| 0.02 | AABB->KDOP26 | 0.0017 | 646.1 B | 0.7 |
| 0.02 | AABB->SupportHull | 0.0017 | 515.0 B | 0.7 |
| 0.02 | AABB->KDOP26->SupportHull | 0.0017 | 777.1 B | 0.9 |
| 0.05 | LinkIAABB | 0.0056 | 384.0 B | 0.5 |
| 0.05 | KDOP26 | 0.0056 | 1.62 KB | 1.3 |
| 0.05 | SupportHull | 0.0056 | 832.0 B | 1.1 |
| 0.05 | AABB->KDOP26 | 0.0056 | 696.0 B | 0.8 |
| 0.05 | AABB->SupportHull | 0.0056 | 540.0 B | 0.7 |
| 0.05 | AABB->KDOP26->SupportHull | 0.0056 | 852.0 B | 1.0 |
| 0.10 | LinkIAABB | 0.0229 | 384.0 B | 0.7 |
| 0.10 | KDOP26 | 0.0229 | 1.62 KB | 1.6 |
| 0.10 | SupportHull | 0.0229 | 832.0 B | 1.1 |
| 0.10 | AABB->KDOP26 | 0.0229 | 758.4 B | 1.0 |
| 0.10 | AABB->SupportHull | 0.0229 | 571.2 B | 0.9 |
| 0.10 | AABB->KDOP26->SupportHull | 0.0229 | 945.6 B | 1.3 |
| 0.20 | LinkIAABB | 0.1255 | 384.0 B | 0.8 |
| 0.20 | KDOP26 | 0.1255 | 1.62 KB | 2.2 |
| 0.20 | SupportHull | 0.1255 | 832.0 B | 1.4 |
| 0.20 | AABB->KDOP26 | 0.1255 | 849.9 B | 1.4 |
| 0.20 | AABB->SupportHull | 0.1255 | 617.0 B | 1.1 |
| 0.20 | AABB->KDOP26->SupportHull | 0.1255 | 1.06 KB | 1.7 |
| 0.30 | LinkIAABB | 0.3678 | 384.0 B | 0.7 |
| 0.30 | KDOP26 | 0.3678 | 1.62 KB | 2.5 |
| 0.30 | SupportHull | 0.3678 | 832.0 B | 2.1 |
| 0.30 | AABB->KDOP26 | 0.3678 | 974.7 B | 1.6 |
| 0.30 | AABB->SupportHull | 0.3678 | 679.4 B | 1.5 |
| 0.30 | AABB->KDOP26->SupportHull | 0.3678 | 1.24 KB | 2.4 |
| 0.50 | LinkIAABB | 1.465 | 384.0 B | 0.9 |
| 0.50 | KDOP26 | 1.465 | 1.62 KB | 2.7 |
| 0.50 | SupportHull | 1.465 | 832.0 B | 3.4 |
| 0.50 | AABB->KDOP26 | 1.465 | 1.18 KB | 2.2 |
| 0.50 | AABB->SupportHull | 1.465 | 793.8 B | 2.5 |
| 0.50 | AABB->KDOP26->SupportHull | 1.465 | 1.58 KB | 3.8 |

*a) Link-envelope chain modes.:* Table II extends the standalone link-envelope microbenchmark beyond independently built LinkIAABB, KDOP26, and SupportHull records. The staged chains run the cheap AABB filter first and invoke KDOP26 or SupportHull only for boxes that survive the earlier stage. Across the 2400-box fixed-width corpus, only 28.3% of boxes reach the tighter second stage, so the AABB–KDOP26 and AABB–SupportHull chains average 1.27 $\mu$s and 1.25 $\mu$s per box, compared with 1.94 $\mu$s for standalone KDOP26 and 1.77 $\mu$s for standalone SupportHull. The full AABB–KDOP26–SupportHull cascade averages 1.85 $\mu$s per box. Table IV compresses the same artifact into one aggregate summary view.

*b) LECT mechanism.:* Table III summarizes the cache semantics. The d18 prewarm materializes only the target leaf layer, storing 524287 evidence records while directly evaluating 262144 leaf records in 20.22 s. Reopening the same cache takes 1.460 s and reuses 262144 cached leaf records without fresh FK above the leaves. This is the paper's main cache-mechanism result: the current implementation does perform persistent evidence replay,

TABLE I
RBF-only warm-versus-cold baselines under the reviewer-safe lifelong-cache protocol. Shelf rows read only the prewarmed external d18 evidence and disable active-database backfill; random-scene rows may persist newly materialized online evidence to the active database.

| Exp. | Scene | Mode | Backfill | Wall s | Plan s | Maint s | Query ms | Audit | Repairs | Ext reuse | Active reuse |
|---|---|---|---|---|---|---|---|---|---|---|---|
| E2 | Shelf+IIWA | Cold | no | 1.820 | 1.799 | 0.0213 | 15.98 | 1/1 | 1 | 0 | 4954 |
| E2 | Shelf+IIWA | Warm d18 | no | 1.812 | 1.791 | 0.0212 | 15.57 | 1/1 | 1 | 3420 | 1809 |
| E2 | Random IIWA (easy) | Cold | yes | 0.1821 | 0.1562 | 0.0259 | 2.300 | 1/1 | 0 | 0 | 3249 |
| E2 | Random IIWA (easy) | Warm d18 | yes | 0.1727 | 0.1467 | 0.0259 | 2.332 | 1/1 | 0 | 3002 | 664 |
| E3 | Shelf+IIWA | Cold | no | 1.587 | 1.553 | 0.0333 | 3.679 | 5/5 | 0 | 0 | 10688 |
| E3 | Shelf+IIWA | Warm d18 | no | 1.588 | 1.555 | 0.0335 | 3.568 | 5/5 | 0 | 7201 | 3946 |

TABLE III
RBF-only Gate 0 and lifelong-cache mechanism status after the no-grid refactor. In E5, FK/link-envelope materialization is charged only on the target leaf layer; parent layers are reconstructed by child-hull propagation during checkpoint.

| Stage | Status | Depth | Leaf FK | Nodes | Evidence | Wall s | Reuse | Cache GiB |
|---|---|---|---|---|---|---|---|---|
| Gate 0 | 11/11 | – | – | – | – | – | – | – |
| E5 prewarm | pass | 18 | 262144 | 524287 | 524287 | 20.17 | 0 | 0.2956 |
| E5 reopen | pass | 18 | 0 | 524287 | 524287 | 1.505 | 262144 | 0.2956 |

TABLE IV
Standalone link-envelope representation summary over the 2400-box fixed-width corpus. The table reports per-box staged materialization cost for LinkIAABB, KDOP26, SupportHull, and the AABB-first chain variants; the escalation column is the fraction of boxes that must run at least one post-AABB stage.

| Rank | Representation | Payload | Build/box us | Escalation % |
|---|---|---|---|---|
| 1 | LinkIAABB | 384.0 B | 0.6809 | – |
| 2 | KDOP26 | 1.62 KB | 1.942 | – |
| 3 | SupportHull | 832.0 B | 1.767 | – |
| 4 | AABB->KDOP26 | 854.8 B | 1.273 | 28.3 |
| 5 | AABB->SupportHull | 619.4 B | 1.254 | 28.3 |
| 6 | AABB->KDOP26->SupportHull | 1.06 KB | 1.847 | 28.3 |

but the reported rows support a canonical-identity cache claim rather than a stronger sector-symmetry reuse claim.

*c) Envelope sweep.:* Table IV gives a compact cross-width summary of the same E2 representation study. LinkIAABB remains the cheapest standalone representation at 0.68 $\mu$s per box and 384 B payload. Standalone KDOP26 and SupportHull increase cost to 1.94 $\mu$s and 1.77 $\mu$s, respectively. The AABB-first chains recover much of that overhead: AABB–KDOP26 averages 1.27 $\mu$s per box at 854.8 B, AABB–SupportHull averages 1.25 $\mu$s at 619.4 B, and all three chain rows escalate beyond the AABB stage on only 28.3% of the boxes in the fixed-width corpus.

### D. Dynamic-Update Capability Study

Table V summarizes the local update procedure on the random-scene edit protocol. It creates two matched edit paths: insertion from easy to medium to hard, and deletion from hard to medium to easy. The warm baseline is measured per transition, not as a cumulative path: the source scene first populates an isolated cache namespace,

TABLE V
Random-scene dynamic obstacle updates on matched easy/medium/hard edit paths. Warm denotes an isolated source-scene prewarm followed by a fresh target-scene rebuild, and speedup denotes the ratio of warm median to incremental median.

| Transition | Incr. (s) | Warm (s) | Speedup |
|---|---|---|---|
| easy->medium | 0.0095 | 0.0902 | 9.493 |
| medium->hard | 0.0364 | 0.0702 | 1.929 |
| hard->medium | 0.0014 | 0.0581 | 41.07 |
| medium->easy | 0.000330 | 0.0464 | 140.36 |

and the reported warm time is then a fresh target-scene `build_coverage` call from that cache.

The incremental update is consistently faster than this cache-warm rebuild baseline on the five-seed IIWA dynamic protocol. Insertion medians are 9.5 ms for easy→medium and 36.4 ms for medium→hard, compared with 90.2 ms and 70.2 ms for warm rebuilds. Deletion is cheaper still because previously accepted boxes remain valid when obstacles are removed: hard→medium takes 1.4 ms and medium→easy takes 0.33 ms, versus 58.1 ms and 46.4 ms warm target rebuilds. This remains a capability study rather than a broad dynamic-scene claim, but it confirms that the explicit forest representation and local regrow interfaces are operational in the current code.

## VII. Conclusion

This paper presented RBF, a link-envelope-guided box-forest method for fast multi-query manipulator planning. Its geometric core maps endpoint interval envelopes to link interval envelopes via capsule Minkowski decomposition and convex-hull generator bounding; an RRT-inspired grower builds a scene-specific box forest into a corridor graph queried by graph search and local refinement; LECT optionally caches kinematic envelope records to reduce repeated materialization cost. The experiments show correct Shelf+IIWA external-evidence replay, a leaf-only cache-prewarm mechanism, a strong LinkIAABB setting with useful AABB-first chain modes, and a working dynamic-update path that outperforms warm rebuild on the reported edit protocol.

*a) Limitations.:* The box-corridor guarantee (theorem 1) is conditional on conservative IFK validation and path containment inside the validated box union; the reported rows do not claim that condition, and the dataset

still does not store corridor-containment fields. The strict audit is an implementation-level acceptance rule rather than a continuous safety proof. The cache mechanism supports persistent evidence replay, but the reported results justify only canonical-identity caching and do not imply that replay produces wall-time speedup in every envelope setting.

*b) Scope and Future Work.:* The empirical scope covers Shelf+IIWA, the cache/update studies, and the balanced random-scene benchmark. Broader generalization studies, stronger cache transfer, and additional cross-planner comparisons remain natural next steps.

## References


[1] R. Deits and R. Tedrake, "Computing large convex regions of obstacle-free space through semidefinite programming," in *Algorithmic Foundations of Robotics XI*, 2015, pp. 109–124.

[2] T. Marcucci, M. Petersen, D. von Wrangel, and R. Tedrake, "Motion planning around obstacles with convex optimization," *Sci. Robot.*, vol. 8, no. 84, 2023.

[3] T. Marcucci, J. Umenberger, P. Parrilo, and R. Tedrake, "Shortest paths in graphs of convex sets," *SIAM J. Optim.*, vol. 34, no. 1, pp. 507–532, 2024.

[4] S. M. LaValle, *Planning Algorithms*. Cambridge University Press, 2006.

[5] ——, "Rapidly-exploring random trees: A new tool for path planning," Iowa State University, Tech. Rep. TR 98-11, 1998.

[6] L. E. Kavraki, P. Švestka, J.-C. Latombe, and M. H. Overmars, "Probabilistic roadmaps for path planning in high-dimensional configuration spaces," *IEEE Trans. Robot. Autom.*, vol. 12, no. 4, pp. 566–580, 1996.

[7] S. Karaman and E. Frazzoli, "Sampling-based algorithms for optimal motion planning," *Int. J. Robot. Res.*, vol. 30, no. 7, pp. 846–894, 2011.

[8] S. Redon, Y. J. Kim, M. C. Lin, and D. Manocha, "Interactive and continuous collision detection for avatars in virtual environments," in *IEEE Virtual Reality Conference*, 2004, pp. 117–124.

[9] S. Redon, M. C. Lin, D. Manocha, and Y. J. Kim, "Fast continuous collision detection for articulated models," *J. Comput. Inf. Sci. Eng.*, vol. 5, no. 2, pp. 126–137, 2005.

[10] X. Zhang, S. Redon, M. Lee, and Y. J. Kim, "Continuous collision detection for articulated models using Taylor models and temporal culling," *ACM Trans. Graph.*, vol. 26, no. 3, p. 15, 2007.

[11] A. Amice, H. Dai, P. Werner, A. Zhang, and R. Tedrake, "Finding and optimizing certified, collision-free regions in configuration space for robot manipulators," in *Algorithmic Foundations of Robotics XV*. Springer, 2022, pp. 328–348.

[12] M. Petersen and R. Tedrake, "Growing convex collision-free regions in configuration space using nonlinear programming," 2023. [Online]. Available: https://arxiv.org/abs/2303.14737

[13] P. Werner, A. Amice, T. Marcucci, D. Rus, and R. Tedrake, "Approximating robot configuration spaces with few convex sets using clique covers of visibility graphs," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 10 359–10 365.

[14] D. Blackmore and M. C. Leu, "Analysis of swept volume via Lie groups and differential equations," *Int. J. Robot. Res.*, vol. 11, no. 6, pp. 516–537, 1992.

[15] K. Abdel-Malek, J. Yang, D. Blackmore, and K. Joy, "Swept volumes: foundation, perspectives, and applications," *Int. J. Shape Model.*, vol. 12, no. 1, pp. 87–127, 2006.

[16] S. Gottschalk, M. C. Lin, and D. Manocha, "OBBTree: A hierarchical structure for rapid interference detection," in *ACM SIGGRAPH*, 1996.

[17] E. Larsen, S. Gottschalk, M. C. Lin, and D. Manocha, "Fast proximity queries with swept sphere volumes," University of North Carolina at Chapel Hill, Tech. Rep. TR99-018, 1999.

[18] E. G. Gilbert, D. W. Johnson, and S. S. Keerthi, "A fast procedure for computing the distance between complex objects in three-dimensional space," *IEEE J. Robot. Autom.*, vol. 4, no. 2, pp. 193–203, 1988.

[19] G. v. d. Bergen, "A fast and robust GJK implementation for collision detection of convex objects," *J. Graph. Tools*, vol. 4, no. 2, pp. 7–25, 1999.

[20] J. Pan, S. Chitta, and D. Manocha, "FCL: A general purpose library for collision and proximity queries," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2012, pp. 3859–3866.

[21] R. E. Moore, R. B. Kearfott, and M. J. Cloud, *Introduction to Interval Analysis*. SIAM, 2009.

[22] L. Jaulin, M. Kieffer, O. Didrit, and E. Walter, *Applied Interval Analysis*. Springer, 2001.

[23] J. M. Snyder, "Interval analysis for computer graphics," in *Proc. ACM SIGGRAPH*, 1992, pp. 121–130.

[24] J.-P. Merlet, "Solving the forward kinematics of a Gough-type parallel manipulator with interval analysis," *Int. J. Robot. Res.*, vol. 23, no. 3, pp. 221–235, 2004.

[25] ——, "Interval analysis for certified numerical solution of problems in robotics," *Int. J. Appl. Math. Comput. Sci.*, vol. 19, no. 3, pp. 399–412, 2009.

[26] F. Schwarzer, M. Saha, and J.-C. Latombe, "Adaptive dynamic collision checking for single and multiple articulated robots in complex environments," *IEEE Trans. Robot.*, vol. 21, no. 3, pp. 338–353, 2005.

[27] J. A. Corrales, F. A. Candelas, and F. Torres, "Safe human–robot interaction based on dynamic sphere-swept line bounding volumes," *Robot. Comput.-Integr. Manuf.*, vol. 27, no. 1, pp. 177–185, 2011.

[28] R. Bohlin and L. E. Kavraki, "Path planning using lazy PRM," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2000, pp. 521–528.

[29] C. M. Dellin and S. S. Srinivasa, "A unifying formalism for shortest path problems with expensive edge evaluations via lazy best-first search over paths with edge selectors," in *International Conference on Automated Planning and Scheduling (ICAPS)*, 2016.

[30] E. Schmerling, L. Janson, and M. Pavone, "Optimal sampling-based motion planning under differential constraints: The drift-less case," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2015.

[31] D. Coleman, I. Sucan, S. Chitta, and N. Correll, "Reducing the barrier to entry of complex robotic software: A MoveIt! case study," *J. Softw. Eng. Robot.*, vol. 5, no. 1, pp. 3–16, 2014.

[32] T. Lozano-Pérez, "Spatial planning: A configuration space approach," *IEEE Trans. Comput.*, vol. C-32, no. 2, pp. 108–120, 1983.

[33] R. A. Brooks and T. Lozano-Pérez, "A subdivision algorithm in configuration space for findpath with rotation," *IEEE Trans. Syst. Man Cybern.*, vol. SMC-15, no. 2, pp. 224–233, 1985.

[34] J. J. Kuffner and S. M. LaValle, "RRT-connect: An efficient approach to single-query path planning," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2000, pp. 995–1001.

[35] J. D. Gammell, S. S. Srinivasa, and T. D. Barfoot, "Batch informed trees (BIT*): Sampling-based optimal planning via the heuristically guided search of implicit random geometric graphs," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2015, pp. 3067–3074.

[36] J. D. Gammell, T. D. Barfoot, and S. S. Srinivasa, "Informed sampling for asymptotically optimal path planning," *IEEE Trans. Robot.*, vol. 34, no. 4, pp. 966–984, 2018.

[37] L. Janson, E. Schmerling, A. Clark, and M. Pavone, "Fast marching tree: A fast marching sampling-based method for optimal motion planning in many dimensions," *Int. J. Robot. Res.*, vol. 34, no. 7, pp. 883–921, 2015.

[38] I. A. Şucan, M. Moll, and L. E. Kavraki, "The open motion planning library," *IEEE Robot. Autom. Mag.*, vol. 19, no. 4, pp. 72–82, 2012.

[39] M. Moll, I. A. Şucan, and L. E. Kavraki, "Benchmarking motion planning algorithms: An extensible infrastructure for analysis and visualization," *IEEE Robot. Autom. Mag.*, vol. 22, no. 3, pp. 96–102, 2015.

[40] N. Ratliff, M. Zucker, J. A. Bagnell, and S. Srinivasa, "CHOMP: Gradient optimization techniques for efficient motion planning," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2009, pp. 489–494.

[41] M. Kalakrishnan, S. Chitta, E. Theodorou, P. Pastor, and S. Schaal, "STOMP: Stochastic trajectory optimization for motion planning," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2011, pp. 4569–4574.
[42] J. Schulman, Y. Duan, J. Ho, A. Lee, I. Awwal, H. Bradlow, J. Pan, S. Patil, K. Goldberg, and P. Abbeel, "Motion planning with sequential convex optimization and convex collision checking," *Int. J. Robot. Res.*, vol. 33, no. 9, pp. 1251–1270, 2014.
[43] M. Mukadam, J. Dong, X. Yan, F. Dellaert, and B. Boots, "Continuous-time gaussian process motion planning via probabilistic inference," *Int. J. Robot. Res.*, vol. 37, no. 11, pp. 1319–1340, 2018.